

Developing Effective Tools for Predictive Analytics and Informed Decisions

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Abstract

By utilizing the statistical analysis, analytics, information processing and business intelligence the business processes are understood and decisions are made aiming to improve profitability. Yet due to the involvement of big data, highly non-linear and multicriteria nature of decision making scenarios in today's governance programs the complex analytics models create significant business, operational and technology risks as well as modeling errors presenting the lack of effective modeling system to governance programs. Consequently the traditional approaches have been reported less useful in proper guiding decision-making communication and in drawing insights from big data. Alternatively here the proposed methodology of integration of data mining, modeling and interactive decision-making is studied as an effective approach where what-if scenarios are evaluated and optimization-based decisions are made.

Introduction

Making decisions based on gut feeling, having wrong assumptions about business models and slow reaction to changes in the market have been the main reasons of missing new opportunities and finally collapse. Increasing the usage of business analytics and modeling techniques as the toolset to help firms compete more effectively, navigate business through the challenging economic tasks, and better satisfy regulatory requirements.

In fact understanding the current performance state of a business process and further improvements on its profitability by making informed decisions would be the two fundamental needs to the success of any financial firm to be satisfied by utilizing business analytics. Yet with increasing the usage of the predictive analytics models in financial industries as a differentiating business capability, the quality of e-governance within the financial firms has been even more increased leading to a better insight into the future improvements. To derive a more detailed understanding of business and process dynamics, many frontier financial institutions in which their success totally depends on the quality of their analytics, characterized their business processes by the predictive analytics models to drive more informed decisions. Analytics models as such aim at understanding business operations and planning future improvements by using systematic, quantitative and data-driven processes linking historical data about the business to models, analysis, and predictions. Such modeling approaches of analytics and further business intelligence in understanding the business operations and processes,

planning future improvements and profitability, forecasting, and predictions, due to the dynamic and unstructured nature of big data, involved in today's most multicriteria decision-making and real-life leadership processes and good governance have become less useful. In order to come up with right assumptions about business models and being able to react rapidly to changes a methodology based on machine learning plus optimization, which is in fact an integration of data mining, business modeling, visualization and interactive decision-making is proposed.

Challenges to predictive analytics

Big data- The effective use of big data is the foundation technology upon which today's firms compete. Big data and advanced business analytics have the potential to quickly deliver competitive advantage to those firms that effectively implement the proper ICT tools. However while enterprises builds up huge data storage nicely, organizations are often discovering that they lack the means to draw insights from their big data, as traditional analytics modeling tools are limited to visualizations. Yet producing the big data technologies to identify the stages, critical measures, outcomes, and actions required for companies to effectively develop big data competency would pose a real challenge to predictive analytics. As in fact big data technologies is the competitive advantage to organizations that rely on data-driven decision-making.

Multicriteria nature- Considering modeling real-life business operations and processes there would be typically multiple conflicting criteria which need to be simultaneously evaluated in today's most decision-making, governance, and leadership processes. This would be a challenge to current multidimensional visualization tools of most analytics models.

Overall because of the involvement of big data, highly non-linear and multicriteria nature of decision making scenarios in today's governance programs the complex analytics models create significant business, operational and technology risks as well as modeling errors presenting the lack of effective modeling system to governance programs.

Research Objectives

Developing an environment for producing the predictive analytics models is fundamental to the effectiveness of business strategies and financial decision-making. A complete environment for creating and managing a multicriteria decision-making model for a good governance function on the basis of big data technologies would increase the value of business strategies and quality of overall enterprise risk management programs. In this context providing a complete environment for creating and managing predictive analytics models in a robust, reliable, automated and integrated way would be one of the main objective of this research. The developed tool not only enables the robust modeling and cutting edge visual-based reporting, it can also accelerate the deployment of predictive models into a financial systems.

Here increasing the speed and robustness of model deployment, assessing the challenges faced by financial institutions in development, production and distribution of analytical models would be the strategic objectives.

Further objectives to developing an effective modeling tool for predictive analytics and making informed decisions would include the effective inserting the capability of handling the big data as well as inclusion the multicriteria modeling expertise. The developed tool would enable organizations to assess their big data and analytics competency, using the baseline to define goals, planing for improvements, prioritizing technology and making user investment decisions, and bring business profit into view. This modeling tool can help uncover critical gaps among business units or between business and ICT groups, thus providing a framework for all the stakeholders to collaborate to advance the organization toward a common goal. Further producing big data technologies in which are being applied against some demanding business imperatives in the governance technology today. Here a high-level overview to big data trends in muticriteria decision making models of governance is provided.

Methodology

The methodology is based on machine learning and optimization influenced by the 5,000-cited research works of professor Battiti and his colleagues [2] at the University of Trento, Italy. This technology has been made affordable by most businesses because of the growing amounts of storage and computational

power available at cheap price so the platform can be deployed as a private cloud, public cloud service or as an embedded engine for specific big data appliances.

Machine learning integrated optimization is well suited for dynamic, big, unstructured data so the big data enterprise requirements could be further satisfied by querying, visualizing, obtaining insight or automated actions. In other words the methodology is about the unification of data mining, modeling and interactive decision-making and continuous innovation process powered by a decision-maker and automated learning. In simpler terms, the system learns from data, and models adapt to changes in business requirements. The methodology would deliver the competitive advantages by providing the ability to evaluate many “what if” scenarios before deciding. Business manager intuition will be combined with updated business data and optimization-based decision processes, so that corrective actions and opportunities can be achieved. The developed software tool progressively learns from the decision-maker about business objectives, and it adapts accordingly.

Applications

Business analysts, investment officers, liquidity managers and organizational leaders all consider the predictive analytics and multicriteria modeling for big data as the differentiating capabilities to the governance programs of their own financial institutions. The developed analytics modeling tool as well as

the traditional means of predictive analytics are indeed the core requirement for success in today's most financial market decision-making. In fact financial institutions rely heavily on quantitative analysis and models in most aspects of financial industries. Predictive analytics to big data and modeling functions are further found in many planning, product, and operational corners of the enterprise e.g. healthcare, transportation, risk management, e-commerce, marketing, social networks, retail credit capacity planning, pricing strategies, predicting consumer behavior, modeling the market strategies, investment portfolios, estimating risks, investigating financial fraud, optimize target market campaigns and advertising, innovation policies, innovation strategies, development, economics and financial policies.

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